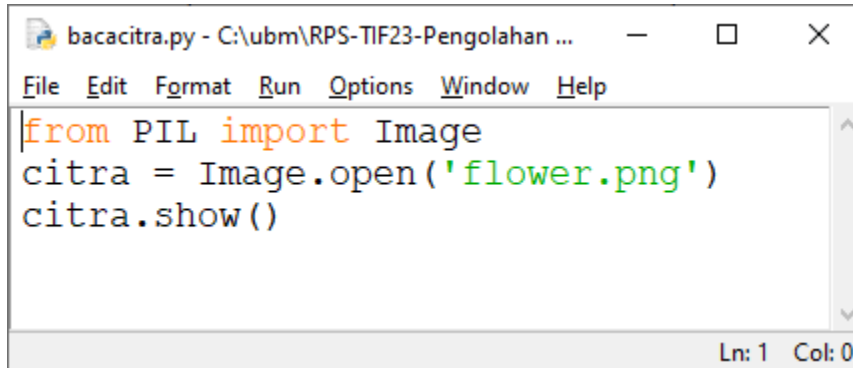


Berikut ini contoh penggunaan pillow untuk membaca dan memisahkan channel RGB

1. Membaca dan menampilkan citra



```
from PIL import Image
citra = Image.open('flower.png')
citra.show()
```

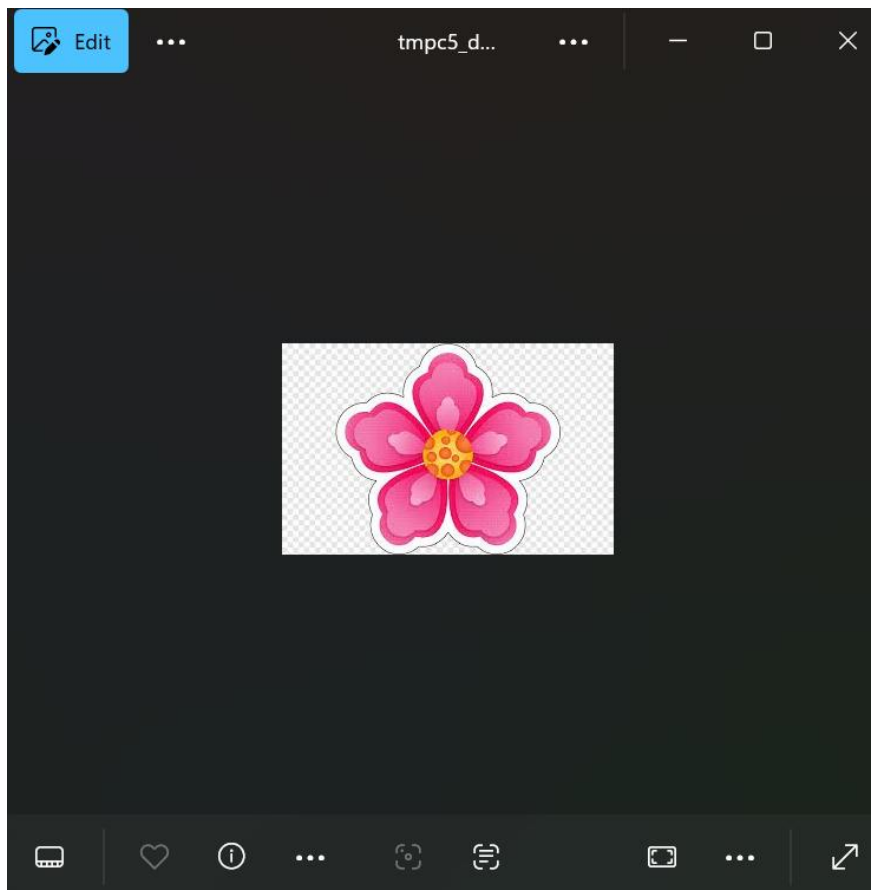
Ln: 1 Col: 0

Ketik dan run program di atas

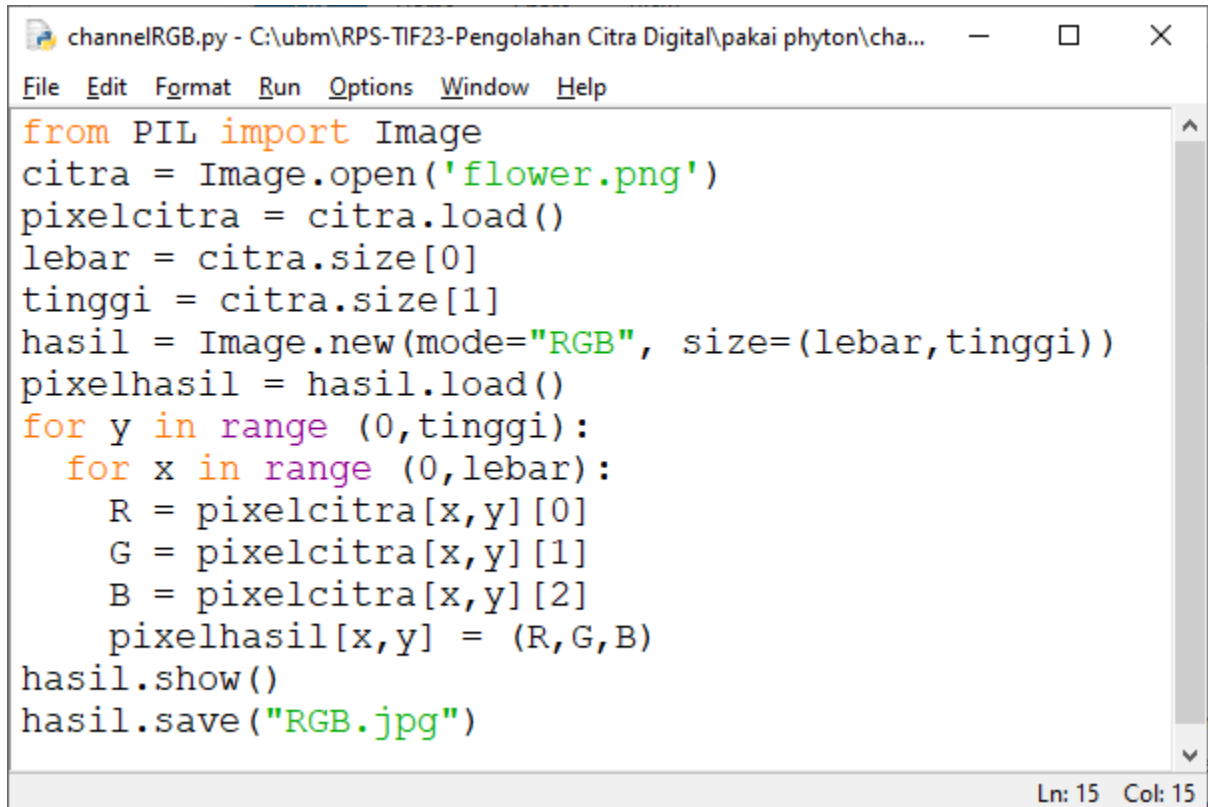
Source Code:

```
from PIL import Image
citra = Image.open('flower.png')
citra.show()
```

Hasil Screenshot:



2. Memisahkan channel R, G dan B



```
channelRGB.py - C:\ubm\RPS-TIF23-Pengolahan Citra Digital\pakai phyton\cha...
File Edit Format Run Options Window Help

from PIL import Image
citra = Image.open('flower.png')
pixelcitra = citra.load()
lebar = citra.size[0]
tinggi = citra.size[1]
hasil = Image.new(mode="RGB", size=(lebar,tinggi))
pixelhasil = hasil.load()
for y in range (0,tinggi):
    for x in range (0,lebar):
        R = pixelcitra[x,y][0]
        G = pixelcitra[x,y][1]
        B = pixelcitra[x,y][2]
        pixelhasil[x,y] = (R,G,B)
hasil.show()
hasil.save("RGB.jpg")

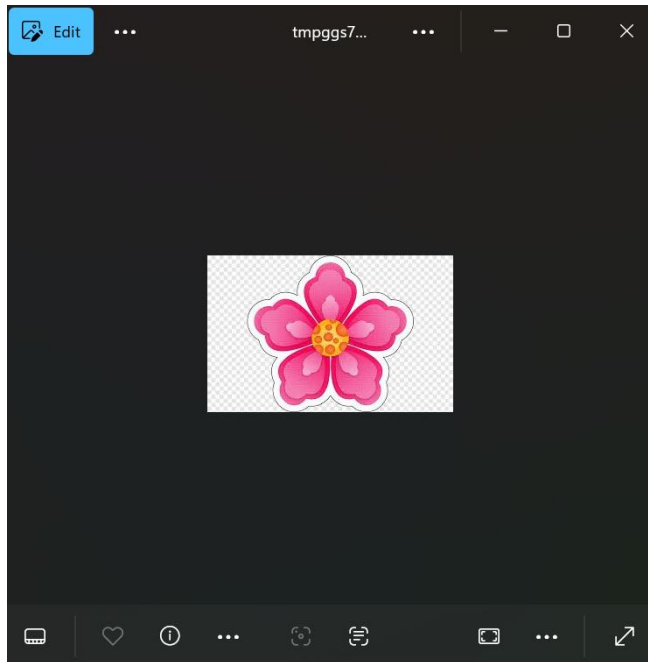
Ln: 15 Col: 15
```

Ketik dan Run Program di atas

Source Code:

```
from PIL import Image
citra = Image.open('flower.png')
pixelcitra = citra.load()
lebar = citra.size[0]
tinggi = citra.size[1]
hasil = Image.new (mode = "RGB", size = (lebar, tinggi))
pixelhasil = hasil.load()
for y in range (0, tinggi):
    for x in range (0, lebar):
        R = pixelcitra[x, y] [0]
        G = pixelcitra[x, y] [1]
        B = pixelcitra[x, y] [2]
        pixelhasil[x, y] = (R, G, B)
hasil.show()
hasil.save("RGB.jpg")
```

Hasil Screenshot:



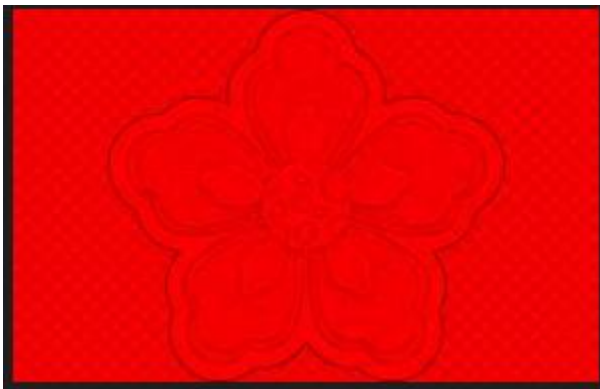
3. Menampilkan channel R saja

```
channelR.py - C:\ubm\RPS-TIF23-Pengolahan Citra Digital\pakai phyton\chan...
File Edit Format Run Options Window Help
from PIL import Image
citra = Image.open('flower.png')
pixelcitra = citra.load()
lebar = citra.size[0]
tinggi = citra.size[1]
hasil = Image.new(mode="RGB", size=(lebar,tinggi))
pixelhasil = hasil.load()
for y in range (0,tinggi):
    for x in range (0,lebar):
        R = pixelcitra[x,y][0]
        G = pixelcitra[x,y][1]
        B = pixelcitra[x,y][2]
        pixelhasil[x,y] = (R,0,0)
hasil.show()
hasil.save("channelR.jpg")
Ln: 1 Col: 0
```

Source Code:

```
from PIL import Image
citra = Image.open('flower.png')
pixelcitra = citra.load()
lebar = citra.size[0]
tinggi = citra.size[1]
hasil = Image.new (mode = "RGB", size = (lebar, tinggi))
pixelhasil = hasil.load()
for y in range (0, tinggi):
    for x in range (0, lebar):
        R = pixelcitra[x, y] [0]
        G = pixelcitra[x, y] [1]
        B = pixelcitra[x, y] [2]
        pixelhasil[x, y] = (R, 0, 0)
hasil.show()
hasil.save("channelR.jpg")
```

Hasil Screenshot:



3a. Buatlah program untuk menampilkan channel G saja

Source Code:

```
from PIL import Image
citra = Image.open('flower.png')
pixelcitra = citra.load()
lebar = citra.size[0]
tinggi = citra.size[1]
hasil = Image.new (mode = "RGB", size = (lebar, tinggi))
pixelhasil = hasil.load()
for y in range (0, tinggi):
    for x in range (0, lebar):
        R = pixelcitra[x, y] [0]
        G = pixelcitra[x, y] [1]
        B = pixelcitra[x, y] [2]
        pixelhasil[x, y] = (0, G, 0)
hasil.show()
```

```
hasil.save("channelG.jpg")
```

Hasil Screenshot:

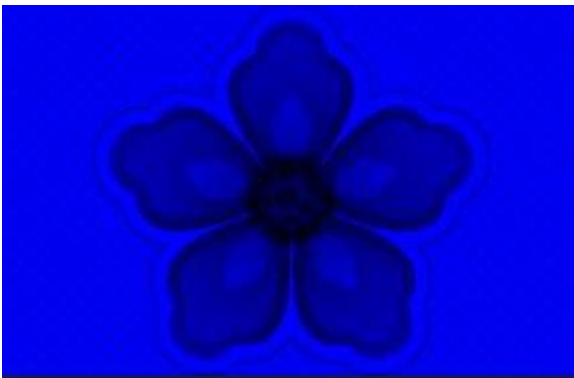


3.b buatlah program untuk menampilkan channel B saja

Source Code:

```
from PIL import Image
citra = Image.open('flower.png')
pixelcitra = citra.load()
lebar = citra.size[0]
tinggi = citra.size[1]
hasil = Image.new (mode = "RGB", size = (lebar, tinggi))
pixelhasil = hasil.load()
for y in range (0, tinggi):
    for x in range (0, lebar):
        R = pixelcitra[x, y] [0]
        G = pixelcitra[x, y] [1]
        B = pixelcitra[x, y] [2]
        pixelhasil[x, y] = (0, 0, B)
hasil.show()
hasil.save("channelB.jpg")
```

Hasil Screenshot:



3.c buatlah program untuk menampilkan channel R dan G

Source Code:

```
from PIL import Image
citra = Image.open('flower.png')
pixelcitra = citra.load()
lebar = citra.size[0]
tinggi = citra.size[1]
hasil = Image.new (mode = "RGB", size = (lebar, tinggi))
pixelhasil = hasil.load()
for y in range (0, tinggi):
    for x in range (0, lebar):
        R = pixelcitra[x, y] [0]
        G = pixelcitra[x, y] [1]
        B = pixelcitra[x, y] [2]
        pixelhasil[x, y] = (R, G, 0)
hasil.show()
hasil.save("channelRnG.jpg")
```

Hasil Screenshot:



3.d buatlah program untuk menampilkan channel G dan B

Source Code:

```
from PIL import Image
citra = Image.open('flower.png')
pixelcitra = citra.load()
lebar = citra.size[0]
tinggi = citra.size[1]
hasil = Image.new (mode = "RGB", size = (lebar, tinggi))
pixelhasil = hasil.load()
for y in range (0, tinggi):
    for x in range (0, lebar):
        R = pixelcitra[x, y] [0]
        G = pixelcitra[x, y] [1]
        B = pixelcitra[x, y] [2]
```

```
        pixelhasil[x, y] = (0, G, B)
hasil.show()
hasil.save("channelGnB.jpg")
```

Hasil Screenshot:



3.e buatlah program untuk menampilkan channel R dan B

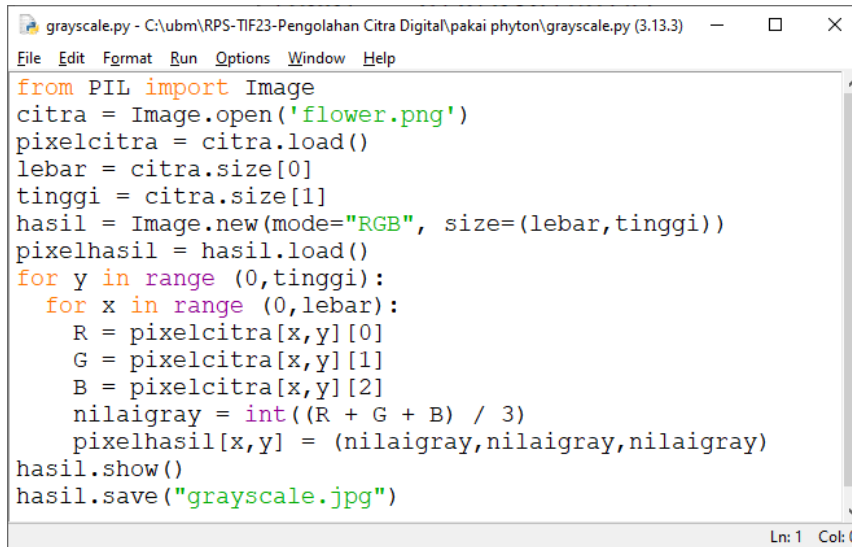
Source Code:

```
from PIL import Image
citra = Image.open('flower.png')
pixelcitra = citra.load()
lebar = citra.size[0]
tinggi = citra.size[1]
hasil = Image.new (mode = "RGB", size = (lebar, tinggi))
pixelhasil = hasil.load()
for y in range (0, tinggi):
    for x in range (0, lebar):
        R = pixelcitra[x, y] [0]
        G = pixelcitra[x, y] [1]
        B = pixelcitra[x, y] [2]
        pixelhasil[x, y] = (R, 0, B)
hasil.show()
hasil.save("channelRnB.jpg")
```

Hasil Screenshot:



4. Citra Grayscale



The screenshot shows a Python IDE window titled 'grayscale.py - C:\ubm\RPS-TIF23-Pengolahan Citra Digital\pakai phyton\grayscale.py (3.13.3)'. The code is as follows:

```
from PIL import Image
citra = Image.open('flower.png')
pixelcitra = citra.load()
lebar = citra.size[0]
tinggi = citra.size[1]
hasil = Image.new(mode="RGB", size=(lebar,tinggi))
pixelhasil = hasil.load()
for y in range (0,tinggi):
    for x in range (0,lebar):
        R = pixelcitra[x,y][0]
        G = pixelcitra[x,y][1]
        B = pixelcitra[x,y][2]
        nilaigray = int((R + G + B) / 3)
        pixelhasil[x,y] = (nilaigray,nilaigray,nilaigray)
hasil.show()
hasil.save("grayscale.jpg")
```

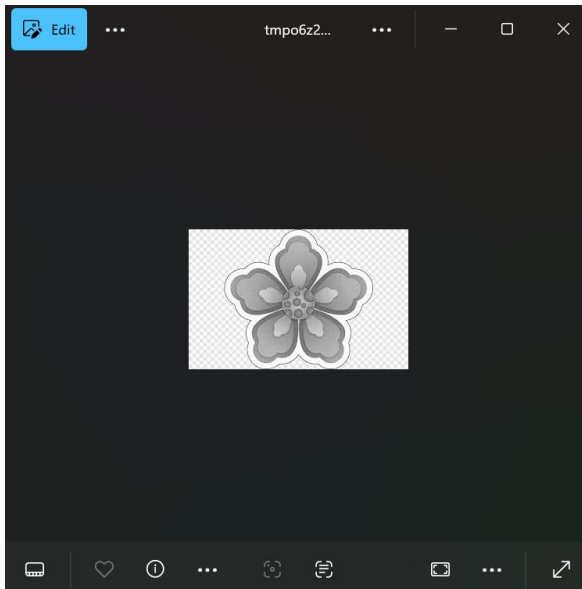
The status bar at the bottom right indicates 'Ln: 1 Col: 0'.

Ketik dan run program di atas

Source Code:

```
from PIL import Image
citra = Image.open('flower.png')
pixelcitra = citra.load()
lebar = citra.size[0]
tinggi = citra.size[1]
hasil = Image.new (mode = "RGB", size = (lebar, tinggi))
pixelhasil = hasil.load()
for y in range (0, tinggi):
    for x in range (0, lebar):
        R = pixelcitra[x, y] [0]
        G = pixelcitra[x, y] [1]
        B = pixelcitra[x, y] [2]
        nilaigray = int ((R + G + B) / 3)
        pixelhasil[x, y] = (nilaigray, nilaigray, nilaigray)
hasil.show()
hasil.save("grayscale.jpg")
```


Hasil Screenshot:



5. Monochrome

```
monochromR.py - C:\ubm\RPS-TIF23-Pengolahan Citra Digital\pakai phyton\...
File Edit Format Run Options Window Help
from PIL import Image
citra = Image.open('flower.png')
pixelcitra = citra.load()
lebar = citra.size[0]
tinggi = citra.size[1]
hasil = Image.new(mode="RGB", size=(lebar,tinggi))
pixelhasil = hasil.load()
for y in range (0,tinggi):
    for x in range (0,lebar):
        R = pixelcitra[x,y][0]
        G = pixelcitra[x,y][1]
        B = pixelcitra[x,y][2]
        nilaigray = int((R + G + B) / 3)
        pixelhasil[x,y] = (nilaigray,0,0)
hasil.show()
hasil.save("monochromR.jpg")
Ln: 1 Col: 0
```

Contoh monochrome R

Source Code:

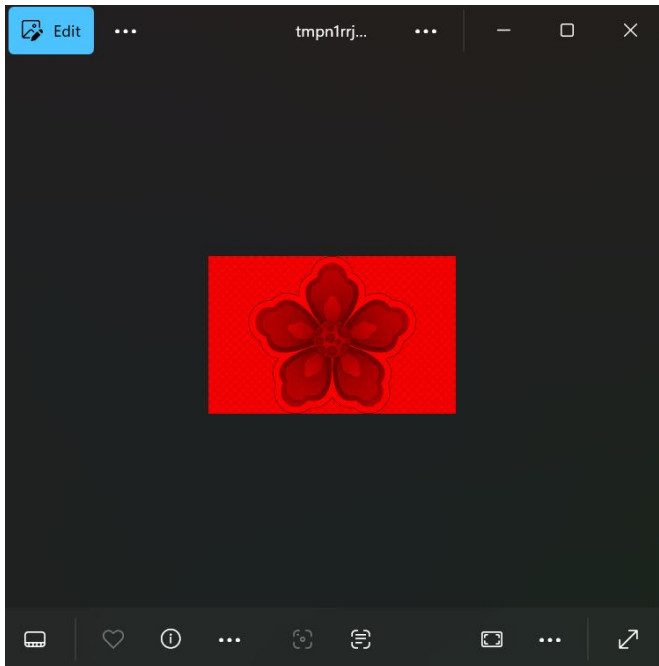
```
from PIL import Image
citra = Image.open('flower.png')
pixelcitra = citra.load()
lebar = citra.size[0]
tinggi = citra.size[1]
hasil = Image.new (mode = "RGB", size = (lebar, tinggi))
pixelhasil = hasil.load()
for y in range (0, tinggi):
```

```

for x in range (0, lebar):
    R = pixelcitra[x, y] [0]
    G = pixelcitra[x, y] [1]
    B = pixelcitra[x, y] [2]
    nilaigray = int ((R + G + B) / 3)
    pixelhasil[x, y] = (nilaigray, 0, 0)
hasil.show()
hasil.save("monochromeR.jpg")

```

Hasil Screenshot:



5.a. buatlah program untuk menampilkan monochrome G

Source Code:

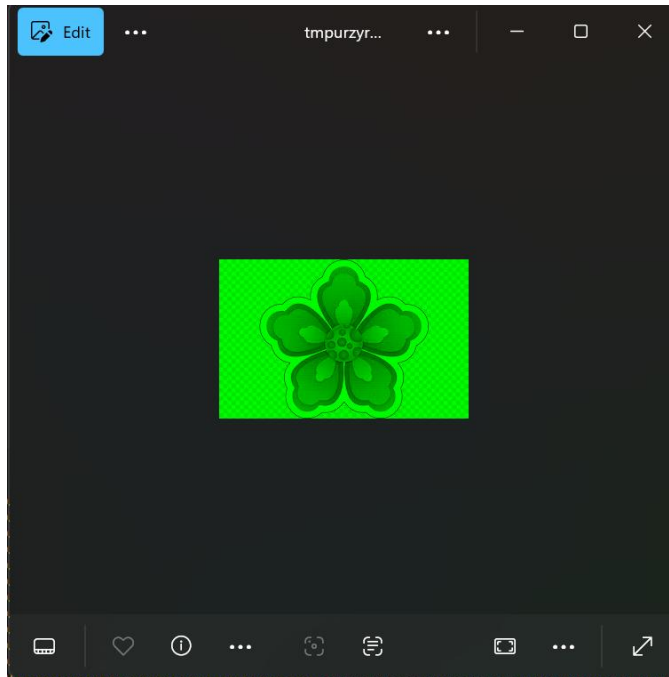
```

from PIL import Image
citra = Image.open('flower.png')
pixelcitra = citra.load()
lebar = citra.size[0]
tinggi = citra.size[1]
hasil = Image.new (mode = "RGB", size = (lebar, tinggi))
pixelhasil = hasil.load()
for y in range (0, tinggi):
    for x in range (0, lebar):
        R = pixelcitra[x, y] [0]
        G = pixelcitra[x, y] [1]
        B = pixelcitra[x, y] [2]
        nilaigray = int ((R + G + B) / 3)
        pixelhasil[x, y] = (0, nilaigray, 0)
hasil.show()

```

```
hasil.save("monochromeG.jpg")
```

Hasil Screenshot:

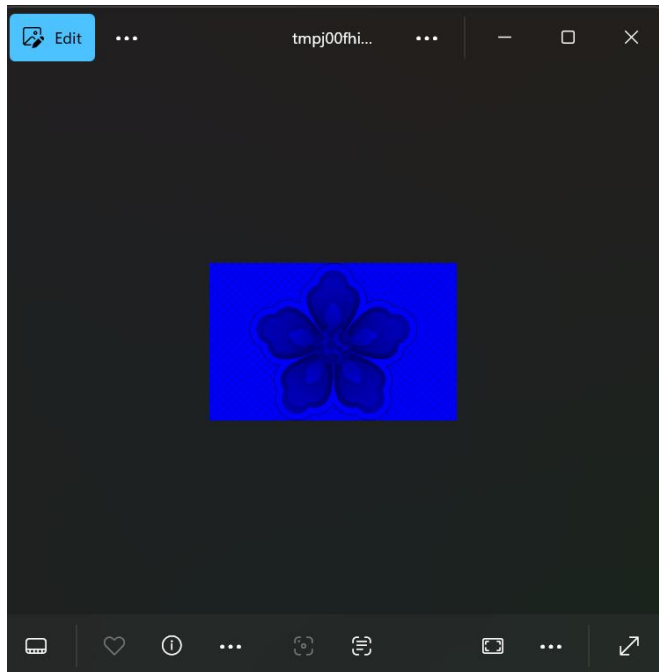


5.b. buatlah program untuk menampilkan monochrome B

Source Code:

```
from PIL import Image
citra = Image.open('flower.png')
pixelcitra = citra.load()
lebar = citra.size[0]
tinggi = citra.size[1]
hasil = Image.new (mode = "RGB", size = (lebar, tinggi))
pixelhasil = hasil.load()
for y in range (0, tinggi):
    for x in range (0, lebar):
        R = pixelcitra[x, y] [0]
        G = pixelcitra[x, y] [1]
        B = pixelcitra[x, y] [2]
        nilaigray = int ((R + G + B) / 3)
        pixelhasil[x, y] = (0, 0, nilaigray)
hasil.show()
hasil.save("monochromeB.jpg")
```

Hasil Screenshot:

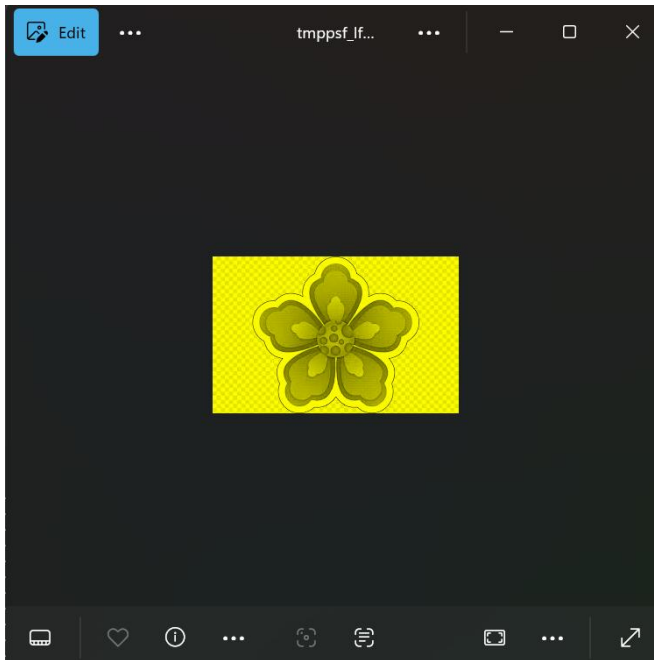


5.c. buatlah program untuk menampilkan monochrome R dan G

Source Code:

```
from PIL import Image
citra = Image.open('flower.png')
pixelcitra = citra.load()
lebar = citra.size[0]
tinggi = citra.size[1]
hasil = Image.new (mode = "RGB", size = (lebar, tinggi))
pixelhasil = hasil.load()
for y in range (0, tinggi):
    for x in range (0, lebar):
        R = pixelcitra[x, y] [0]
        G = pixelcitra[x, y] [1]
        B = pixelcitra[x, y] [2]
        nilaigray = int ((R + G + B) / 3)
        pixelhasil[x, y] = (nilaigray, nilaigray, 0)
hasil.show()
hasil.save ("monochromeRnG.jpg")
```

Hasil Screenshot:

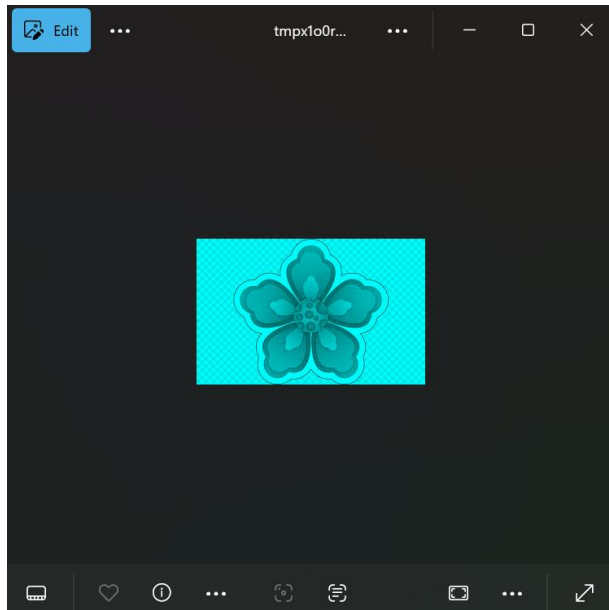


5.d. buatlah program untuk menampilkan monochrome G dan B

Source Code:

```
from PIL import Image
citra = Image.open('flower.png')
pixelcitra = citra.load()
lebar = citra.size[0]
tinggi = citra.size[1]
hasil = Image.new (mode = "RGB", size = (lebar, tinggi))
pixelhasil = hasil.load()
for y in range (0, tinggi):
    for x in range (0, lebar):
        R = pixelcitra[x, y] [0]
        G = pixelcitra[x, y] [1]
        B = pixelcitra[x, y] [2]
        nilaigray = int ((R + G + B) / 3)
        pixelhasil[x, y] = (0, nilaigray, nilaigray)
hasil.show()
hasil.save("monochromeGnB.jpg")
```

Hasil Screenshot:



5.e. buatlah program untuk menampilkan monochrome R dan B

Source Code:

```
from PIL import Image
citra = Image.open('flower.png')
pixelcitra = citra.load()
lebar = citra.size[0]
tinggi = citra.size[1]
hasil = Image.new (mode = "RGB", size = (lebar, tinggi))
pixelhasil = hasil.load()
for y in range (0, tinggi):
    for x in range (0, lebar):
        R = pixelcitra[x, y] [0]
        G = pixelcitra[x, y] [1]
        B = pixelcitra[x, y] [2]
        nilaigray = int ((R + G + B) / 3)
        pixelhasil[x, y] = (nilaigray, 0, nilaigray)
hasil.show()
hasil.save ("monochromeRnB.jpg")
```

Hasil Screenshot:

